

Varying Status Quo Conditions in Stated Preference Surveys: Empirical Evidence for Water Quality

Abstract

This study examines how current conditions shape people’s preferences for public policies that improve environmental quality. We use a stated preference survey with experimentally varied status quo levels in a setting with a spatially explicit freshwater health score across a wide geographic area. We provide causal evidence that people have a higher willingness to pay when status quo water quality levels are worse. These findings highlight the importance of accurately describing the status quo condition in stated preference studies. This study also provides empirical insights into the monetized economic benefits of landscape-wide water quality improvements in Canada. We find that the public’s willingness to pay for water quality improvements varies by spatial scale and proximity to the policy site. Local improvements in home basins or sub-basins are estimated to have a higher willingness to pay than distant ones. Recreational users have substantially higher willingness to pay, underscoring the joint role of non-use and use values.

3.2. Introduction

Monetized benefits of water quality improvements play a crucial role in shaping environmental policies, particularly as the degradation of freshwater ecosystems has led policy makers to prioritize their management (UN-Water 2021; Naidoo and Adamowicz 2005; Keiser, Kling, and Shapiro 2019). These benefits underpin cost-benefit analyses by quantifying trade-offs and net economic value to society, incorporating both market and nonmarket ecosystem services (Goldstein et al. 2012; Naidoo and Adamowicz 2005; Arrow et al. 1996). Stated-preference (SP) surveys, which estimate people’s willingness to pay (WTP) for environmental improvements, are a cornerstone of this valuation (Johnston et al. 2017). The SP literature is continuously evolving to address the critical nuances of policy relevance and the complexities of regulatory decision-making (Johnston et al. 2017; Vossler et al. 2023; Moore et al. 2023; Bateman et al. 2023).

SP practitioners often face difficult decisions when describing status quo (SQ) conditions to respondents. In SP surveys, the SQ alternative provides a baseline description of current or future conditions and acts as a comparison for the policy alternatives. In many contexts and particularly water quality SP surveys, there is often limited data for accurately defining the SQ water quality condition. For

example, limited data may suggest that water quality is either “poor” or “moderate” in an area and the researcher needs to decide which water quality to use as a SQ in a SP survey. Do these researcher decisions regarding the SQ description affect elicited preferences and values for environmental changes?

To study this question we employ a split-sample approach to isolate the effect of status quo conditions, thereby providing a robust assessment of how public preferences and welfare measures vary with the status quo presented to respondents. To do so, we conducted a SP survey administered to 4,000 households across the Canadian prairie provinces of Alberta, Saskatchewan, and Manitoba. Our approach builds on recent advancements in SP research, including improved identification of ecosystem services, refined water quality metrics, and the incorporation of spatial dimensions through interactive visual maps—all designed to align with regulatory objectives (Moore et al. 2023; Kling, Phaneuf, and Zhao 2012; Johnston et al. 2017; Vossler et al. 2023; Bateman et al. 2023).

We employed a four-version split-sample experimental design, with each version incorporating a distinct, experimentally varied SQ water quality level. Respondents were randomly assigned to one of four survey versions, each featuring a unique SQ water quality level based on the World Wildlife Fund’s (WWF) freshwater health score, adapted for Canada (WWF-Canada 2020). While choice attributes and proposed improvements remained consistent across versions, the SQ was manipulated to vary respondents’ baseline perceptions. This design enables us to isolate the causal impact of SQ on the utility of water quality improvements, overcoming the challenges associated with relying solely on observational variability of SQ across river basins. In particular using observed SQ introduces endogeneity, as actual water quality is often correlated with unobserved factors, such as socioeconomic conditions, land use, pollution sources, or differential exposure, meaning people in poorer-quality areas may have different priorities or less experience with high-quality water. In contrast, experimentally controlled SQ allows us to isolate causal effects by randomly assigning baseline water quality levels, thereby mitigating the endogeneity and perceptual biases inherent in observational data. This experimental control provides a robust framework for identifying how SQ water quality levels affect elicited preferences and values for water quality changes.

We find that people’s preferences for water quality improvements varies by spatial scale and proximity to the improvement. The annual household marginal WTP for an improvement in the water quality score is estimated to be 44% higher for local basins (\$226) than for non-local basins (\$156). For sub-basin-level improvements, marginal WTP is \$147 for local sub-basins and \$86 for non-local sub-basins. These findings highlight the importance of spatial context in shaping public preferences for

water quality interventions, informing more effective and equitable environmental policies in Canada.

Our results indicate that public preferences for water quality improvements vary systematically with the SQ water quality conditions. By leveraging experimental variation in SQ, we find robust empirical evidence that researchers' decisions regarding how the SQ water quality level is framed meaningfully influence revealed preferences in SP valuation. Specifically, poorer SQ conditions amplify the utility associated with improvements, resulting in up to a 52% higher willingness to pay for equivalent improvements when the baseline is severely degraded compared to when it is relatively healthy, particularly for improvements at the local sub-basin scale. These findings highlight the importance of accurate definition of baseline water quality in SP studies, as it systematically influences public preferences for water quality improvements and consequently affects policy-relevant valuations and environmental decision-making.

This study advances the SP literature on water quality valuation by systematically examining how SQ water quality conditions shape public preferences, an underexplored area in prior research. The limited number of studies that have explored this issue have focused on the changing nature of the SQ conditions and whether environmental quality is improving, remaining the same, or decreasing under the SQ. For instance, Maguire et al. (2018) used a three-way split-sample design with valuation questions that incorporated a constant, a decline, or an improvement in future conditions without policy interventions across survey versions. Their findings reveal that WTP varies with future SQ trajectories, with marginal WTP increasing from constant to declining to improving conditions, highlighting reference dependence in shaping policy support. A related stream of the SP literature has elicited perceptions of SQ water quality and compared responses to researcher-defined SQ conditions, finding that the SQ-perceived group exhibited stronger preferences and higher WTP for water quality improvements (Marsh, Mkwara, and Scarpa 2011). This underscores the critical influence of perceived SQ on environmental valuation outcomes, although self-reported baseline SQ risks endogeneity with preferences. In contrast, our study uses a split-sample design that exogenously manipulates current SQ water quality across respondents, avoiding reliance on future projections or subjective reports. This experimental control enables a more rigorous test of SQ effects, controlling for endogeneity concerns that arise from relying on observational variability of SQ. It also offers empirical insights into the role of SQ conditions in preference formation, with critical implications for reducing uncertainty in regulatory analysis and improving the design of environmental valuation surveys.

This study also contributes to the limited Canadian SP literature on valuing

water quality, where only a few studies have addressed the complexities of regional and policy-relevant contexts (Macaskill and Lloyd-Smith 2022). Compared to the U.S., progress in Canada has been slower in aligning SP applications with regulatory decision-making needs. We advance this literature by employing a comprehensive water quality index that captures the interplay of human activities, pollution, and ecological integrity, producing more robust WTP estimates for realistic policy contexts (Arrow et al. 1996; Carson and Mitchell 1993; Moore et al. 2023). By integrating detailed water quality measures, spatial heterogeneity, and rigorous survey design, this study strengthens the accuracy and policy applicability of SP literature on water quality valuation in Canada.

3.3. Study Area

In this study, we focus on four major river basins within the Prairie Provinces (Alberta, Saskatchewan, Manitoba), spanning over 800,000 square kilometers, which are further divided into 18 sub-basins (Figure 3.1). These watersheds are critical to agriculture and the economy, providing irrigation for extensive farmland, potable water for urban centers, and essential ecosystem services such as water purification and flood mitigation (Schindler and Donahue 2006). Consequently, the region faces significant ecological threats, particularly from agricultural runoff, nutrient loading, and habitat fragmentation, which pose risks to water quality and ecosystem resilience. Climate change, land-use changes, and potential water scarcity further amplify the ecological and economic costs of degradation (Western Economic Diversification Canada 2020; Schindler and Donahue 2006). Empirical evidence on monetized economic benefits and ecological integrity remains a critical component of policy decision-making, given the region’s economic dependence, complex trans-boundary nature, ecological vulnerabilities, and climate-driven pressures for managing water resources sustainably (Warren, Lulham, and Lemmen 2022).

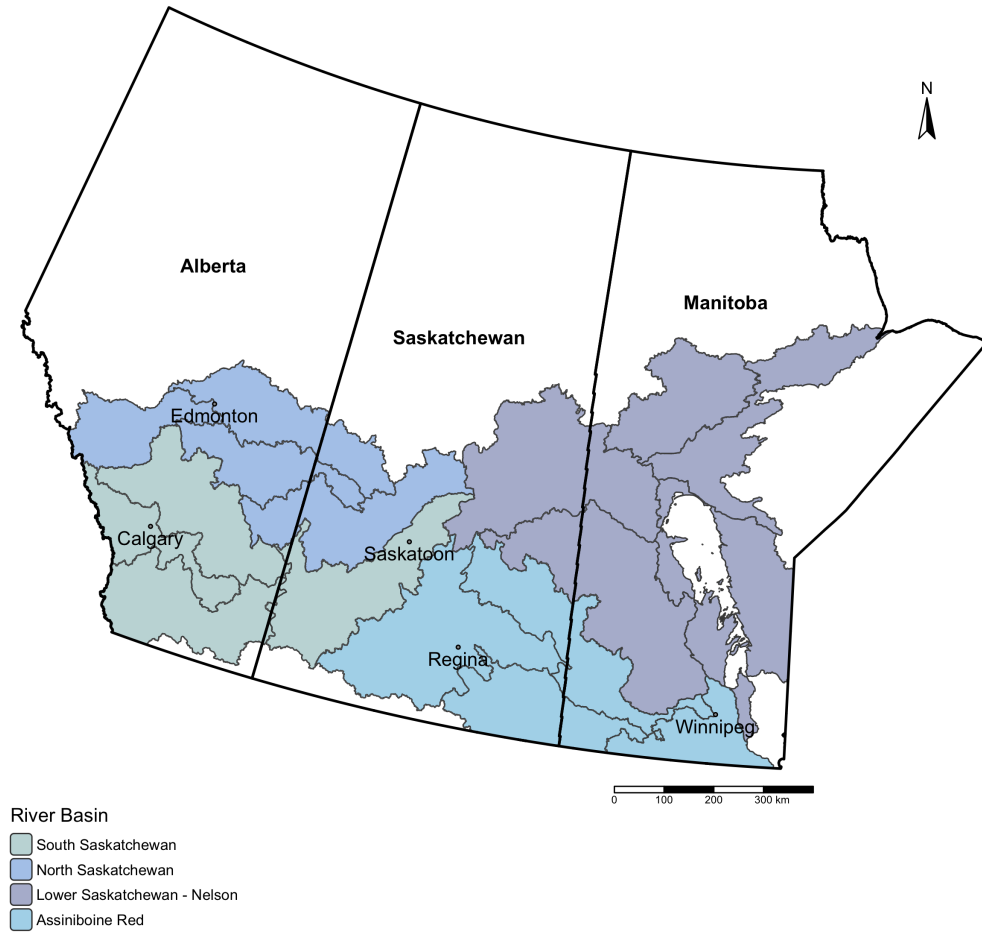


Figure 3.1: The study area includes four main river basins across the Prairie provinces in Canada.

Notes: Our study area consists of four river basins, namely the North Saskatchewan, South Saskatchewan, Assiniboine-Red, and Lower Saskatchewan-Nelson River basins, which are further divided into 18 river sub-basins.

3.4. Stated Preference Survey

3.4.1. Survey Structure

We built on Vossler et al. (2023) and tailored the survey instrument to the Canadian context. In particular the survey included four distinct sections, and the full

survey instrument is provided in the supplementary materials. The online survey encouraged respondents to use a laptop, desktop, or large tablet for optimal viewing of graphics. This was followed by an authentication screen requesting the first three digits of the respondent’s residential postal code to verify their location and proceed with the survey.¹ Introductory screens outlined the study’s purpose, researcher affiliations, and government funding through the Global Water Futures program.

In Section 1, respondents provided opinions on water resources and conditions near their homes. Section 2 used visual and written descriptions to characterize variations in water appearance, natural flow, and biodiversity. To verify image comprehension and act as attention checks, respondents selected the image depicting the best water quality from a set of three. Incorrect answers triggered a refresher on image interpretation. These images were paired with detailed variations corresponding to the WWF’s five-level health score, with lower numerical values assigned to more naturally functioning ecosystems, as shown in Figure 3.2. After reviewing water condition levels, respondents viewed a map showing the spatial distribution of water quality for a non-local basin and identified the water quality level of a specific city within it. Using a map of their local basin, they identified the water quality of their local basin. Section 3 included valuation scenarios, detailed in the next section. It concluded with follow-up questions assessing whether respondents’ votes aligned with incentive-compatible elicitation, which attributes influenced their choices, their response uncertainty, beliefs about the tax burden’s distributional aspects, and general thoughts on the water quality proposals. Section 4 was brief and primarily focused on water-based recreation, including detailed trip information for 2024.

3.4.2. Characterizing Water Quality and Status Quo for Respondents

Water quality is complex to measure, and SP practitioners often face challenges in conveying water quality and the SQ to respondents. We use the overall freshwater health scores, derived from the WWF Canada Freshwater Assessment framework, to define water quality as a commodity for which we measure preferences (WWF-Canada 2020). This national-scale framework measures deviations in ecological integrity from a water body’s natural condition, provides a standard, scientifically grounded method for evaluating health across Canadian river basin and sub-basins based on the international best practices (Boulton 1999; Roux 1999; Norris and Thoms 1999; Birk et al. 2012; U.S. Environmental Protection Agency 2013) and

¹Since only the first three postal code digits were collected, some Forward Sortation Areas (FSAs) spanned sub-basin borders. For those respondents, we included questions with a map showing sub-basin distribution across the province, asking them to select their local sub-basin.

domestic initiatives (Kidd et al. 2011; Alberta Environment and Sustainable Resource Development 2012; Maaskant and Quinlan 2012). The overall health score is a composite index based on four core metrics: hydrology, water quality, benthic macroinvertebrates, and fish, which represent key components of aquatic ecosystems and are widely monitored in Canada (WWF-Canada 2020). Each metric is assessed through quantitative indicators that measure current status and, where sufficient data exist, temporal trends, with data sourced from long-term monitoring by federal, provincial, and regional entities, including Environment and Climate Change Canada, provincial governments, community groups, and Indigenous-led initiatives². We used the health scores published during the 2020 assessment, which were the most recent available at the time of survey development ³.

The WWF’s freshwater health score is attractive for our purposes, as it is designed to provide comparable interpretations of biological health across watersheds, offering spatial transferability and grounding in scientifically credible ecological principles. We use these spatially explicit health scores to carefully define water quality as an environmental good and service to be valued, ensure consistent interpretation across respondents, and generate empirical estimates that are broadly comparable across space and linked to a policy-relevant metric of ecological integrity (Vossler et al. 2023). The overall health score is categorized into five levels: very poor, poor, fair, good, and very good, with each level reflecting a distinct degree of change from baseline ecosystem function and integrity at the sub-watershed scale. The scale of the scores captures the extent of impacts from various stressors resulting from human activities in watersheds. The status of water quality can therefore be improved by actions undertaken to address anthropogenic stressors, such as enhancing natural water flow and controlling nutrient loading from agricultural land use (Chambers et al. 2012; Vossler et al. 2023).

We translated the ecological concepts underlying each health score into both visual and textual representations, ensuring they are understandable to survey respondents in order to elicit economic values associated with water quality improvements. In particular, we characterize the appearance of a natural stream by highlighting visual features such as water color and clarity, physical modifications that alter the stream’s natural flow, the condition of riparian zones and banks, and the diversity

²More details on WWF’s technical protocols for health assessment, the derivation of the overall water quality index, and the specific data sources are available in the Technical Protocol report at <https://watershedreports.wwf.ca>.

³We also reviewed the most recent health scores published in 2025. When comparing the health scores used in our survey with the most recently published data, we found no significant changes between the two assessments. Most importantly, none of the missing health scores have been updated in the latest edition

of fish, benthic macroinvertebrates, and aquatic plants supported by varying levels of health score. These are important drivers or are associated with the core metrics upon which the overall health score is built, and are therefore likely to be associated with different health scores and serve as key determinants of the ecosystem conditions of natural streams. We summarized this information along with the varying visual features corresponding to each of the five levels of the overall health score. Each level visually depicts the appearance of a natural stream or river, accompanied by a snapshot of biological diversity illustrated through a gradient of benthic macroinvertebrates and aquatic plants. The visual and associated verbal narratives corresponding to five level of health score define the water quality as commodity (Figure 3.2).

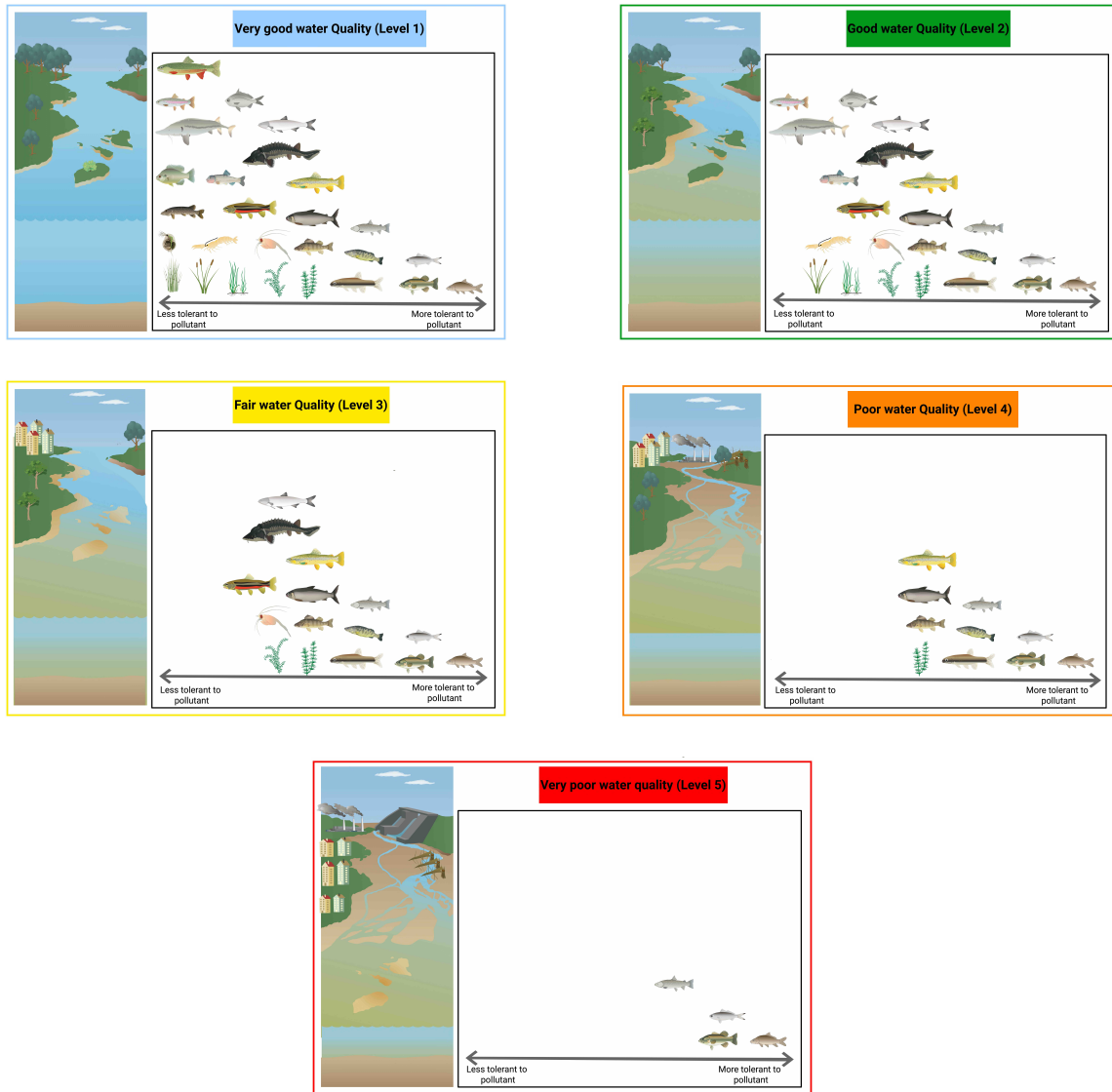


Figure 3.2: Visual representation of a stream at five corresponding WWF's overall health score.

Notes: We develop visual representations for each level of WWF's overall health score, depicting the underlying core metrics and accounting for their variation at each level, to facilitate comparable interpretations of biological health across watersheds.

To incorporate the spatial dimension of water quality improvements, we assigned spatially explicit baseline health score values across all sub-basins in our study area

(Figure 3.1). At the time of data acquisition, some sub-basins lacked health score values due to data gaps. For instance, among 18 sub-basins across four basins within the study area, eight sub-basins did not have an overall health score. To address this issue, we imputed health scores for sub-basins with missing data based on the overall health scores of neighboring sub-watersheds, their flow directions, and the threat levels assigned to the sub-watersheds with missing data.⁴ To account for the uncertainty associated with this imputation, we considered two possible scenarios of the assigned health scores: in one scenario, we assigned a higher health score, and in the other, we assigned a lower health score for the missing cases.

The overall health scores are based on the WWF Freshwater Assessment in 2021, which was the most recent health score available at the time of the study. Since we propose water quality improvements for a future time span, we adjusted the current baseline health scores based on assessed threat levels to account for potential recent and future changes in water quality. For these threat-based adjustments, we considered two baseline scenarios. In the first scenario, we downgraded the health score by two units if the threat level was assessed as high or very high, and by one unit if the threat level was assessed as moderate or low. In the second scenario, we downgraded the health score by one unit only if the threat level was assessed as very high; all other cases remained unchanged.

In total, this resulted in four potential baseline health score scenarios for each sub-basin. For sub-basins with missing overall health scores, variation stemmed from both the initial imputation of health scores and the threat-based adjustment. For sub-basins with available overall health scores, variation resulted solely from the two threat-level adjustment scenarios.⁵ Across the four baseline conditions, 5.6% of the study area had a baseline health score classified as “very good”, 66.7% had scores ranging from “good” to “fair”, 16.7% had scores ranging from “fair” to “poor”, and 11.1% had scores classified as “very poor”, based on the WWF freshwater health score classification, as shown in Figure 3.3.

⁴We used the WWF Freshwater Assessment Report to identify threat levels at the sub-basin level. The framework incorporates key indicators to capture current stressors in Canada’s freshwater systems.

⁵The threat-based adjustments were nested within the initial imputation of missing health score values, resulting in four baseline conditions.

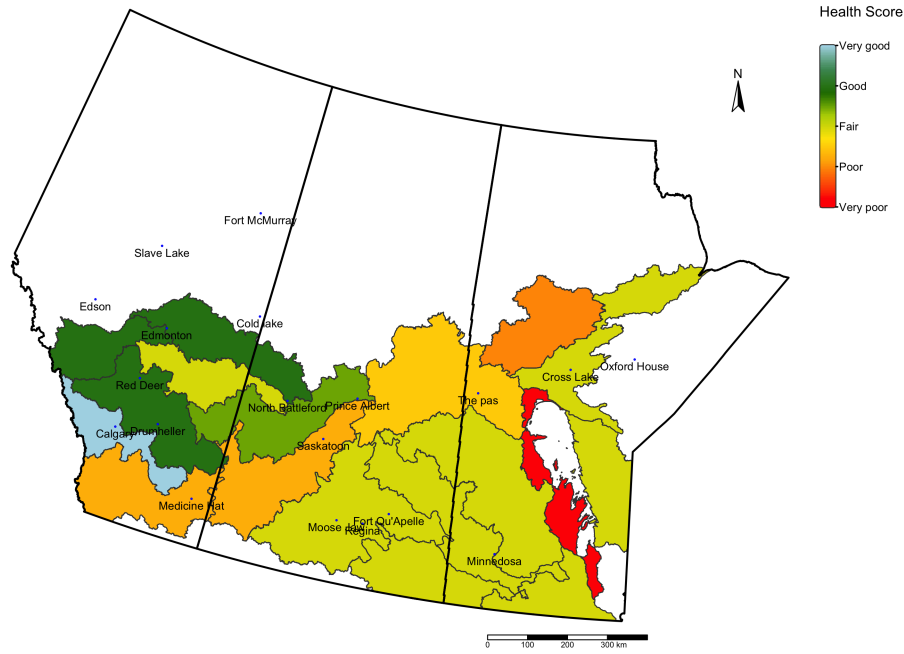


Figure 3.3: The average baseline Health Score for 18 sub-basins under four baseline conditions across survey versions.

Notes: We use four baseline conditions based on threat levels and missing health scores. Here, we depict the average health score of each sub-basin across the four baseline conditions.

3.4.3. Valuation Questions

The valuation exercise was designed to empirically estimate the degree to which people prefer ecosystems closer to their natural state. To estimate households' WTP for improvements in water quality, the survey presented up to seven valuation scenarios tailored to each respondent. We followed established survey best practices to encourage truthful responses and to interpret answers to the valuation scenarios as reflections of real economic trade-offs. Drawing on the literature on incentive-compatible survey elicitation (Carson and Groves 2007; Vossler, Doyon, and Rondeau 2012), we framed the valuation scenarios as advisory referenda, employed a coercive payment mechanism, and instructed respondents to consider each referendum independently

(Vossler et al. 2023). Furthermore, we emphasized the survey’s consequentiality by informing participants that it was government-funded and that the results could influence public policy decisions.

The survey included valuation scenarios defined by four attributes: the spatial scale (basin or sub-basin), whether the change occurred in a local or non-local area, the water health score change, and a cost attribute (Table 3.1). We use experimental variation to examine how economic welfare responds to two spatial dimensions, based on four different levels of improvement in the health score. The first spatial dimension is the scale of water quality changes, where we propose improvements at either the full basin level or only within a sub-basin. The second spatial dimension is the location of the changes, which has two levels: whether the changes occur within the respondent’s local basin or not. The tax increase was defined as a household-level annual tax, randomly assigned and unavoidable if the referendum was approved, applied over five years period. To reduce respondent cognitive burden, we grouped the scenarios into spatial unit blocks and water health score change blocks. Within the spatial unit blocks, we defined two spatial scales: an entire river basin and sub-basin. Within a spatial unit block, we held the spatial scale fixed while varying the health score change scenario (four possibilities). In a health score change block, we held the health score change scenario fixed while varying the spatial unit. For each respondent, we assigned either two spatial unit blocks or two health score change blocks, with one block representing their local basin or sub-basin and the other representing a non-local basin or sub-basin. We determined each respondent’s local sub-basin based on their postal code provided at the beginning of the survey, which was then matched to both their local and non-local sub-basins. By presenting scenarios that either include or exclude the respondent’s local basin, we are able to differentiate the economic values of nearby improvements and how these values vary with distance.

Table 3.1: Attributes and levels of the choice scenario

Attribute	Level
Health score changes	1. Improve health score to level 1 2. Improve health score to level 2 3. Improve health score to level 3 4. Improve health score level by 1 unit
Spatial scale of changes	1. Full basin 2. Only a sub-basin
Location of changes	1. Changes within the local basin 2. Changes within the non-local basin
Cost (Annual tax increase over five years)	\$25, \$75, \$100, \$150, \$200, \$250, \$350, \$500, \$750

A valuation scenario consisted of maps displaying the policy area, along with details showing the baseline and proposed improvements in water quality. A table was used to summarize the areas under different health score levels within the policy basin under both current and improved conditions. The scenario also included a summary table presenting the average changes in health score across the policy basin, the size of the policy area, the household cost, and a vote solicitation framed as a public referendum (Figure 3.4).

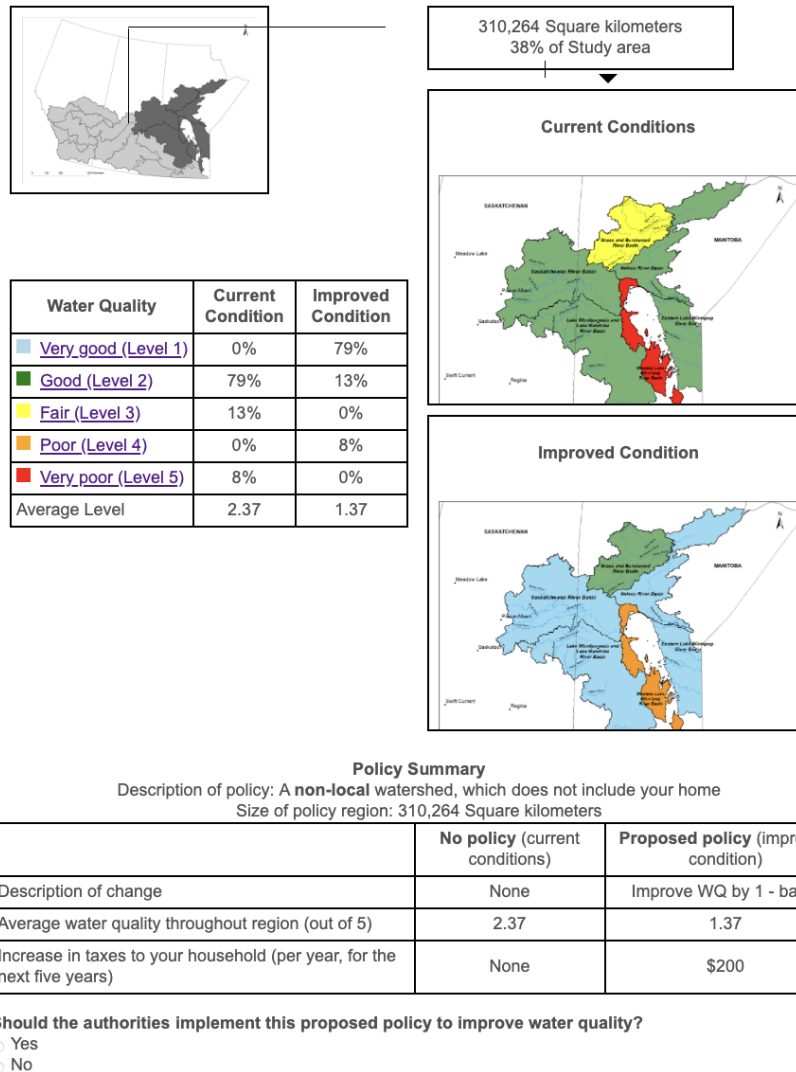


Figure 3.4: A sample referendum scenario

Notes: Choice scenario includes a visual representation of maps showing the policy area, with color-coded depictions of current and improved conditions, along with a table summarizing the area under each health score. At the end, a summary table presents key policy details, including the size of the policy area, the proposed change, how the average health score changes before and after the policy, the average cost per household, and finally, the referendum question.

3.4.4. Survey Testing

We used the Voxco survey engine to code the survey for completion by an online panel. The experimental design and survey functionality were tested in July 2024 through two online focus groups, each with 8–9 participants, to ensure effective operation and clear materials and questions. Participants lived in the study area, were recruited using random digit dialing, and were invited to attend one of the potential dates. Drawing on focus group insights, we conducted a pilot survey with 1,000 respondents across three provinces in December 2024 using the revised survey instrument to verify functionality and gather preliminary results, which informed further refinements to the final survey. Additionally, the pilot survey data were used to estimate preliminary models, confirming that our experimental design worked. Based on these insights, we adjusted the bid levels while keeping the rest of the experimental design unchanged.

3.4.5. Survey Administration

The final survey instrument was administered to a sample of 3,850 people residing in our study area, verified by postal codes, between July 2025 and August 2025⁶. The survey was conducted in partnership with the Canadian Hub for Applied and Social Research (CHASR) using CHASR’s online probability-based panel. We used an opt-in internet panel to survey the general population in the Prairie provinces living in the study area, using quota-based sampling. With opt-in internet panels, participants may be more likely to complete a survey if they are interested in the topic, raising concerns about non-response bias (Johnston et al., 2017). To reduce potential self-selection bias, the survey invitation email omitted information about the survey topic. A quota-based sampling strategy ensured the sample was representative of province of residence, age, and gender. Furthermore, we used appropriate weighting for respondents to align with the 2021 census population targets.

3.5. Econometric Framework

The theoretical framework for the econometric analysis of data from our discrete choice experiment is Random Utility Maximization (RUM) theory, which assumes that the utility an individual derives depends on both the observable attributes of

⁶Though our data collection spanned three months, the majority of responses were collected during the early phase of the online survey. A growing literature examines whether public preferences are sensitive to temporal factors such as seasonality and survey timing. Assessing these temporal dynamics represents a potential extension of our research.

the choices and unobserved factors affecting well-being, the latter captured through a random component (McFadden 1974). Let the utility of respondent i derived from option j in voting scenario t be given by:

$$U_{ijt} = \beta_{program} + \alpha C_{ijt} + \beta_{wq}^{sl} \Delta WQ_{ijt}^{sl} + \beta' X_{ijt} + e_{ijt} \quad (1)$$

where utility U_{ijt} is the utility that individual i derives from alternative j in choice situation t , and is assumed to be additively separable into the program-specific constant for the water quality improvement program $\beta_{program}$, cost C_{ijt} (presented to respondents as an increase in household taxes), water quality changes for spatial unit sl , with $s \in \{\text{Basin, Sub-basin}\}$ and $l \in \{\text{Local, Non-local}\}$ and a vector of other attributes X_{ijt} . The vector X_{ijt} includes individual-specific covariates invariant across alternatives j and thus cancels in the utility difference, not affecting choice probabilities. α , β_{wq}^{sl} and β are the estimable parameters and e_{ijt} is a random error term assumed to be independently and identically distributed Type I extreme value.

First, we estimate Equation 1 using a binary logit (BL) model specification (Train 2009). Second, we use random parameter logit (RPL) models that allows utility parameters to be random and follow parametric distributions (Revelt and Train 1998). The RPL approach accommodates multiple sources of unobserved variability, including heterogeneity in tastes across respondents, and it also accounts for panel data from repeated choice tasks. In particular, we model preferences for water quality improvements across varying spatial units and the program-specific constants as random, using a normal probability density function, the most common specification for characterizing preference heterogeneity (Train 2009).⁷

3.5.1. Welfare Measures

Within this estimation framework, the negative of the parameter on the cost variable, $-\alpha$, measures the marginal utility of income. The parameter on the water quality score, β_{wq}^{sl} , represents the marginal utility of a change in water quality at a given spatial scale (sl). Since an increase in the water quality score corresponds to a decline in water quality, the estimated β_{wq}^{sl} is negative, reflecting the disutility of worsening water quality. Consequently, the negative of this parameter, $-\beta_{wq}^{sl}$, measures the marginal utility of a one-unit improvement in water quality (i.e., a one-unit decrease in the score). Dividing the marginal utility of a one-unit decrease in the water

⁷All models were estimated using the Apollo package in R (Hess and Palma 2019). To estimate the random parameter logit (RPL) models, we used simulated maximum likelihood estimation with 5,000 Sobol draws.

quality score by the marginal utility of income yields the marginal WTP for a one-unit improvement in water quality at different spatial scales, as follows:

$$WTP_{\text{marginal}}^{sl} = \frac{-\beta_{\text{wq}}^{sl}}{-\alpha} \quad (2)$$

The total WTP (WTP_{total}^{sl}) measures the monetary value of a specific water quality improvement scenario compared to the current condition. This is calculated by dividing the difference in utility between the improved and current scenarios by the marginal utility of income. The utility difference accounts for the change in the water quality score and the program-specific constant (β_{program}), which captures the baseline utility of the improved scenario relative to the status quo. The total WTP is given by:

$$WTP_{\text{total}}^{sl} = -\frac{\beta_{\text{program}} + (WQ_{\text{improved}}^{sl} - WQ_{\text{current}}^{sl}) * \beta_{\text{wq}}^{sl}}{\alpha} \quad (3)$$

We apply the delta method to calculate standard errors for the welfare measures, ensuring robust inference for the WTP estimates derived from Model 2 (Train 2009).

3.6. Results

The results section is organized into four subsections. First, we present findings from the primary models estimating preferences for water quality improvements and discuss the associated welfare estimates. Second, we extend the modeling framework to explore the effect of status quo water quality conditions on shaping public preferences for water quality enhancements. Then, we examine model specifications that evaluate how spatial attributes and respondents' recreational behavior influence public preferences for water quality improvement programs. Finally, we assessed survey response validity to ensure reliable data and conducted sensitivity analyses to evaluate result robustness.

3.6.1. Modeling Preferences for Water Quality Improvements

Table 3.2 reports the choice model parameter estimates for the water quality data. Model 1 uses the Binary Logit (BL) specification, which assumes homogeneous preferences across individuals. The program-specific constant for the policy alternative is positive and significant, indicating a baseline preference for policy interventions over the status quo. The cost coefficient is negative and significant, consistent with expected disutility from increased costs. We interacted water quality scores, specific

to the choice scenarios, with indicators for the spatial scale of the policy area (i.e., basin or sub-basin), further distinguished as local or non-local based on respondents' home locations. As higher water quality score values indicate poorer water quality, the estimated negative coefficients suggest that respondents derive lower utility from deteriorating conditions and prefer improved water quality, as expected.

In Model 2, we employed a RPL model, which incorporates random parameters to capture unobserved preference heterogeneity⁸. The estimated mean for the program-specific constant is positive and the standard deviation parameter is estimated to be large suggesting diverse preferences for policy interventions. The estimated cost coefficient is negative and statistically significant, as expected, consistent with the negative marginal utility of price. Results indicate that individuals are approximately 1.5 times more sensitive to changes in water quality within the local basin compared to the non-local basin. Improvements at the sub-basin level also show a preference for local sub-basins over non-local ones, although marginal utility gains are smaller compared to basin-level improvements.

⁸This model is estimated in the preference space with a fixed cost coefficient. As shown by Train and Weeks (2005), when the cost coefficient is fixed, the model is mathematically equivalent to one specified in WTP space. For completeness and to facilitate interpretation, we also present the results from the Model 2 specification estimated directly in WTP space. These results are reported in Appendix Table A1.

Table 3.2: Estimation results for the primary model specifications, using Binary Logit (BL) and Random Parameters Logit (RPL) models.

	Model 1	Model 2	
		Mean	Std. Dev
Program Constant	0.931*** (0.0475)	1.554*** (0.148)	2.422*** (0.118)
Cost	-0.003*** (1e-04)	-0.005*** (3e-04)	
Δ Water Quality Score			
Basin:local	-0.396*** (0.039)	-1.1*** (0.178)	0.835*** (0.241)
Basin:non-local	-0.361*** (0.0367)	-0.758*** (0.128)	0.347 (0.32)
Sub Basin:local	-0.28*** (0.0265)	-0.715*** (0.089)	0.528*** (0.118)
Sub Basin:non-local	-0.18*** (0.0236)	-0.418*** (0.069)	0.417*** (0.118)
Individuals	3,825		3,825
Voting Sceneario	26,848		26,848
Log-Likelihood at solution	-16,269		-13,133
McFadden's R^2	0.060		0.239
BIC	32,599		26,377

Both models use covariates that interact WWF's freshwater health scores with indicators for the spatial scale of the policy area (basin or sub-basin), further distinguished as local or non-local based on respondents' home locations, to capture variation in marginal utility from water quality improvements depending on locality and spatial scale. Model 1, estimated as a Binary Logit (BL) model, assumes homogeneous preferences across individuals. Model 2, a Random Parameter Logit (RPL) model, accommodates panel data from repeated choice tasks, and substantially improves model fit. Robust standard errors are reported in the parenthesis. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates from our model specification allow us to examine how welfare is affected by the spatial scale and the respondent's proximity to the location of the water quality improvement, defined as the additional WTP for expanding a

policy area, as well as by whether the policy site is local or non-local. Figure 3.5 presents annual marginal WTP estimates for water quality improvements over five years, based on Model 2, across varying spatial scales⁹. The marginal WTP for improvements at the basin level, including the respondent’s sub-basin, is \$226. This amount decreases to \$156 when the policy site excludes the respondent’s local basin. Scaling down to the sub-basin level results in lower WTP, with \$147 for local sub-basins and \$86 for non-local sub-basins. However, because each basin consists of four to six sub-basins, comparing WTP on a per-square kilometer basis indicates higher values at the sub-basin scale relative to the basin scale. Households consistently express higher WTP for improvements within their local sub-basins compared to non-local ones. These findings suggest diminishing marginal WTP as the distance between the respondent’s residence and the improved area increases.

⁹Here, in Model 2, the use of the normal distribution for random parameters offers flexibility but allows for potentially implausible negative WTP values. Following best practices (Hensher and Greene 2003), we conduct a robustness check using lognormal distributions in WTP space. The results confirm our core findings, while yielding relatively higher, right-skewed WTP estimates, which may be attributed to the long upper tail of the lognormal distribution (Appendix Table A2.

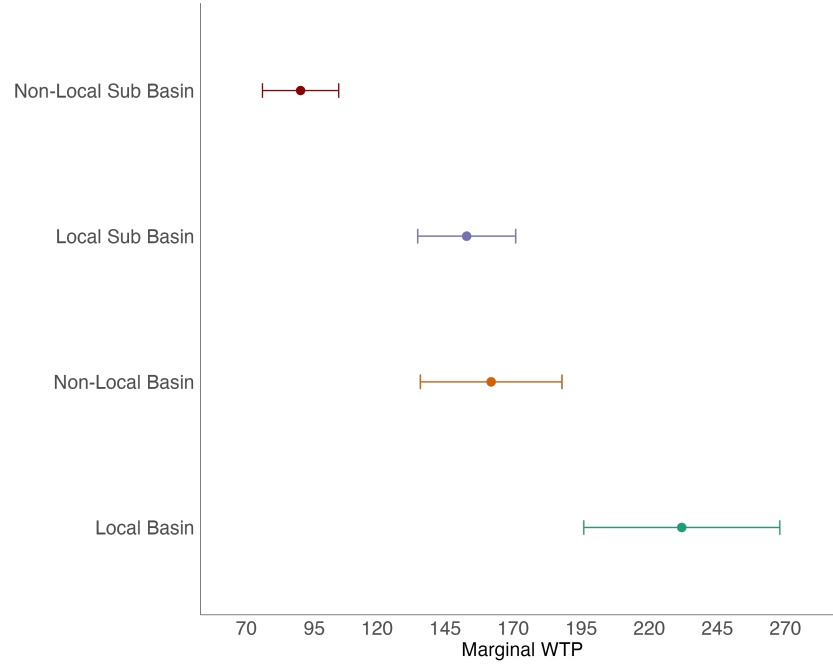


Figure 3.5: Marginal willingness to pay for improvement in water quality

Notes: The figure indicates the marginal household WTP (in 2024 dollars) per year over a five-year period for a policy that aims to improve water quality at varying spatial scales and localities. A local policy refers to one that improves water quality at the basin or sub-basin scale where the household resides, whereas a non-local policy does not include the household’s resident basin. Estimates are derived from Model 2, as described in Table 2.

Using Model 2 results, we estimate the total WTP for improving the water quality score from each sub-basin’s current water quality level to a “good” level (Level 2) at the sub-basin scale, based solely on local values. Figure 3.6 depicts the spatial distribution of local households’ total WTP for achieving a “good” health score from current water quality conditions, with estimated values ranging from \$320 to \$995 and higher WTP in areas with lower baseline health scores in our study region.

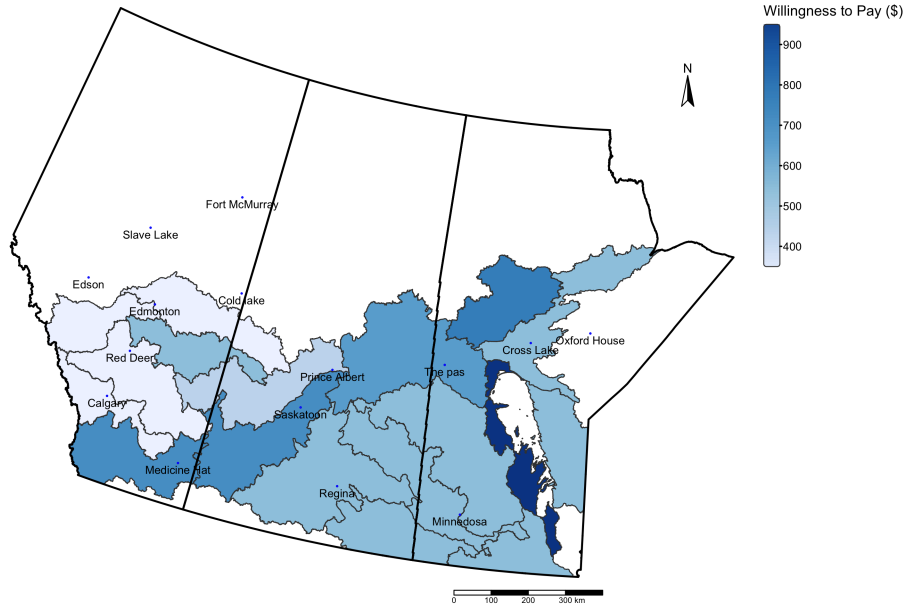


Figure 3.6: Spatial distribution of total WTP for improving the health score to Level 2. (\$ per household in the affected watershed, annual payment for 5 years).

Notes: Total WTP values are calculated at the local sub-basin level for a scenario of improving the health score to a “good” condition (Level 2) from the current health score, based on the average across four baseline conditions.

3.6.2. Impact of Status Quo Water Quality Conditions

Table 3.3 summarizes the estimation results from the model examining how SQ water quality conditions influence public preferences for water quality improvement programs. In our experimental design, variation in SQ water quality arises not only from inherent differences across sub-basins but also from experimentally induced variation across survey versions within those sub-basins. Our objective is to assess whether this exogenous, experimentally assigned variation in SQ affects preferences for water quality improvements, thereby addressing the endogeneity concerns that arise when relying solely on observational differences in SQ across locations. To do so, we evaluate in Model 3 how the experimentally varied SQ water quality level influences the

utility of choosing a water quality improvement program. Specifically, we include a variable capturing the current water quality score presented in each choice task, and we incorporate basin and sub-basin-level fixed effects to control for both observable SQ differences and unobserved basin-specific factors. This specification allows us to isolate how public preferences respond to exogenous variation in SQ, thereby mitigating endogeneity bias associated with non-random spatial variation in water quality conditions.

The statistically significant positive coefficients for the experimentally varied SQ water quality level indicate that as the experimentally induced SQ worsens, the utility derived from improving water quality increases. This suggests that respondents place a higher value on improvements when the experimentally manipulated baseline conditions are poorer. For example, the total WTP for a one unit improvement in water quality at the Upper South Saskatchewan sub basin, which is one of the sub basins within the South Saskatchewan River Basin, is estimated to be about \$635 when the SQ water quality is at level 5, whereas it is only about \$417 for the same improvement when the SQ is at level 2 (see Figure 3.7).

The results indicate a pronounced framing effect, whereby respondents anchor their preferences to the SQ water quality level. By leveraging experimentally induced variation along with basin- and sub-basin-level fixed effects, we isolate the influence of SQ conditions from unobserved spatial confounders, yielding more credible estimates than those based solely on observational differences. Importantly, the evidence shows that the way researchers define the SQ water quality level can meaningfully shape elicited preferences. Thus, the choice of SQ framing in stated preference surveys is not merely a design detail; it directly influences estimated willingness to pay for water quality improvements. Careful attention to baseline conditions is therefore essential for producing reliable and policy-relevant valuation evidence.

Table 3.3: Estimation Results for the Impact of Baseline Water Quality Conditions on Policy Preferences

Parameter	Model 3	
	Mean	Std. Dev.
Program Constant	0.722** (0.358)	2.42*** (0.12)
Cost	-0.005*** (3e-04)	
Current WQ Level	0.354*** (0.0948)	
Δ Water Quality Score:		
Basin:local	-0.432** (0.198)	0.48 (0.323)
Basin:non-local	-0.244* (0.14)	0.154 (0.136)
Sub Basin:local	-0.6*** (0.103)	0.505*** (0.139)
Sub Basin:non-local	-0.425*** (0.086)	0.401*** (0.125)
Basin/Sub Basin FE	Y	
Individuals	3,825	
Voting Scenario	26,848	
Log-Likelihood at solution	-13,018	
McFadden's R2	0.244	
BIC	26,372	

Notes: This table reports results from Model 3 evaluating how experimentally varied status quo (SQ) water quality influences preferences for water quality improvements. The specification includes a variable indicating the WWF Freshwater Health Index score at the SQ for each choice task, along with fixed effects at the basin and sub-basin levels. This approach captures how the utility of choosing water quality improvement policies is affected by experimentally induced variability in SQ water quality across survey versions, addressing endogeneity concerns that arise from relying on observed variability in SQ water quality levels. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

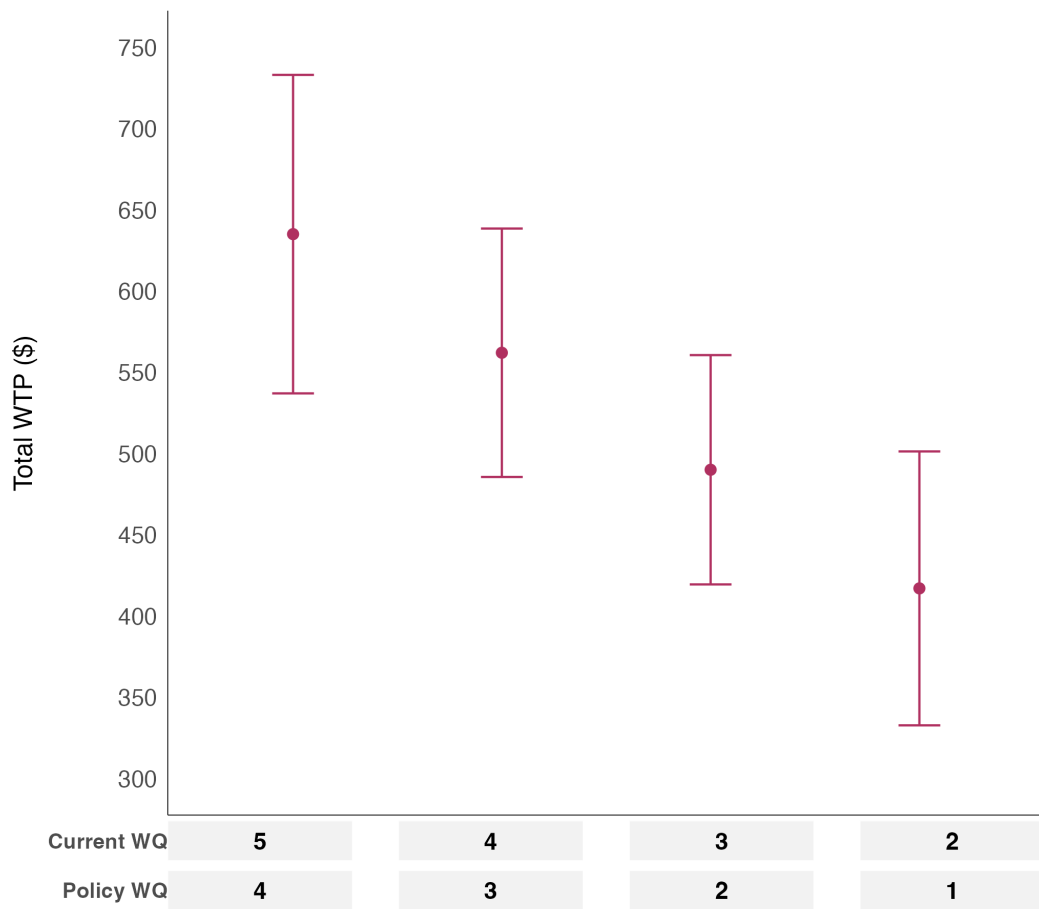


Figure 3.7: Total WTP across varying baseline water quality conditions

Notes: Total willingness to pay for local water quality improvements in the reference sub basin, that is, the Upper South Saskatchewan sub basin, which is a sub basin of the South Saskatchewan River Basin, is estimated using Model 3 parameters, incorporating experimental variation in the status quo based on WWF Freshwater Health Scores.

One concern with the initial imputation of WWF’s overall health score, which is used to overcome missing WQ scores and thread-based adjustments to account for temporal changes in baseline WQ, is the extent to which it aligns with respondents’ perceptions of the SQ water quality level in their local basin. To evaluate this alignment, we included a gauging question that elicited respondents’ perceptions of the presented SQ in their local choice scenarios. Specifically, 59% of respondents indicated that the given baseline WQ level was approximately what they expected, 31% believed it was worse than expected, and 10% perceived it as better than expected. Furthermore, to assess the robustness of our findings to respondents’ perceived SQ, we estimated a model specification that included dummy variables identifying respondents who reported that the presented SQ baseline deviated from the WQ baseline they perceived in their local basin. The results of this specification are summarised in Appendix Table A3. We find no statistically significant difference in utility parameters for respondents who believed the presented SQ water quality level was worse than they expected. However, we find marginally significant positive utility associated with enrolment in the policy programme among respondents who believed the presented SQ baseline WQ was better than they expected.

3.6.3. Influence of Spatial Attributes and Recreational Behavior

In Table 3.4, we present estimates from models that include additional covariates to examine variation in preferences for spatial attributes and respondents’ recreational behavior. Model 4 evaluates whether respondents place an additional premium on non-local choices that are adjacent to their local basin or sub-basin. To do so, we include variables that indicate whether water quality improvements in non-local basins or sub-basins are adjacent to the respondent’s local basin or sub-basin. The results indicate that individuals derive higher marginal utility from water quality improvements in non-local basins that are adjacent to their local basin, compared to non-local basins that are farther away. For instance, there is approximately a 35% additional premium on the marginal WTP for non-local basin-level improvements that are adjacent to the respondent’s local basin (\$169), relative to similar improvements in non-local basins that are not adjacent (\$126).

As freshwater management policies and strategies vary across provinces, in Model 5, we assess how public preferences are shaped by provincial boundaries. Specifically, we use a continuous variable representing the percentage of the policy area within the respondent’s home province. The positive coefficient for this variable suggests that, as the proportion of the policy area within the respondent’s home province increases, the utility of the policy alternative also increases and, this relationship is

statistically significant.

Model 6 assesses how respondents' policy preferences vary based on their recreational behavior. We include a variable indicating whether a respondent made a water-based recreational trip in 2024 to the same basin or sub-basin that is the focus of their choice scenarios. The positive and statistically significant coefficient suggests that respondents who took recreational trips within the policy-relevant basin or sub-basin derive significantly higher utility from the policy alternatives, reflecting greater use values compared to respondents who did not visit water bodies within the policy areas.

Table 3.4: Estimation Results for the Influence of Spatial Attributes and Recreational Behavior on Preferences and Willingness to Pay

Parameter	Model 4		Model 5		Model 6	
	Mean	Std.Dev	Mean	Std.Dev.	Mean	Std.Dev
Program Constant	1.551*** (0.149)	2.426*** (0.119)	1.433*** (0.157)	2.422*** (0.118)	1.551*** (0.142)	2.308*** (0.107)
Cost	-0.005*** (3e-04)		-0.005*** (3e-04)		-0.005*** (2e-04)	
Home Province Share of Policy Site			0.309*** (0.1199)			
Recreational Trip to Choice Sub/Basin					0.475** (0.2189)	
Δ Water Quality Score:						
Basin:local	-1.117*** (0.188)	0.847*** (0.262)	-1.067*** (0.183)	0.813*** (0.264)	-0.909*** (0.164)	0.663*** (0.234)
Basin:non-local	-0.614*** (0.155)	0.31 (0.254)	-0.8*** (0.13)	0.409 (0.288)	-0.654*** (0.118)	0.164 (0.533)
Sub Basin:local	-0.724*** (0.09)	0.527*** (0.115)	-0.674*** (0.09)	0.516*** (0.117)	-0.606*** (0.105)	0.013 (0.031)
Sub Basin:non-local	-0.414*** (0.071)	0.426*** (0.113)	-0.444*** (0.071)	0.418*** (0.112)	-0.441*** (0.066)	0.28** (0.119)
Interact with						
X Non-Local Choice	-0.21* (0.127)	0.392** (0.163)				
Adjacent to Local Sub/Basin						
Individuals		3,825		3,825		3,825
Voting Sceneario		26,848		26,848		26,848
Log-Likelihood at solution		-13,126		-13,119		-13,207
McFadden's R^2		0.239		0.240		0.235
BIC		26,385		26,361		26,537

Notes: This table presents estimation results from models examining how spatial attributes and recreational behavior affect respondents' preferences and willingness to pay (WTP) for water quality improvements, based on WWF's freshwater health score. Results include coefficients from Model 4: assessing price premiums for adjacency to local basins or sub-basins, Model 5: evaluating the effect of the policy area's proportion within the respondent's home province, and Model 6: analyzing the impact of recreational trips to policy basins or sub-basins. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.6.4. Evaluation of Response Validity and Robustness

To evaluate the validity, reliability, and respondent engagement of our stated preference survey, we incorporated several follow-up questions. Responses to questions assessing beliefs related to incentive-compatible elicitation revealed robust and valid preferences. Specifically, 81% of participants reported voting as if their household would incur the stated policy cost, 80% believed the proposed policies would achieve the stated water quality improvements, and 78% responded as if the survey results would influence policymakers. The attributes of the water quality improvement program were salient to respondents, with 91% indicating that their votes were influenced by the level of water quality improvement, 70% by the spatial extent of the policy's impact, and 70% by improvements to water quality near their residence. The survey demonstrated strong perceived neutrality, with 78% of respondents reporting that it allowed them to form their own opinions and 85% expressing confidence in their votes, comparable to voting in a public election.

A sensitivity analysis was conducted using a series of validity questions embedded in the survey. Figure 3.8 illustrates the marginal WTP for a one-unit improvement in water quality score, derived from Model 2 estimates on restricted samples that exclude respondents based on various combinations of validity criteria. These criteria, detailed in the figure notes, encompass protest behavior, lack of engagement, uncertainty in choices, disbelief in survey consequentiality, and inadequate consideration of proposal attributes. The results demonstrate robustness across the full sample, though some criteria have excluded notable respondents from the dataset, indicating that the findings are not driven by responses exhibiting validity concerns.

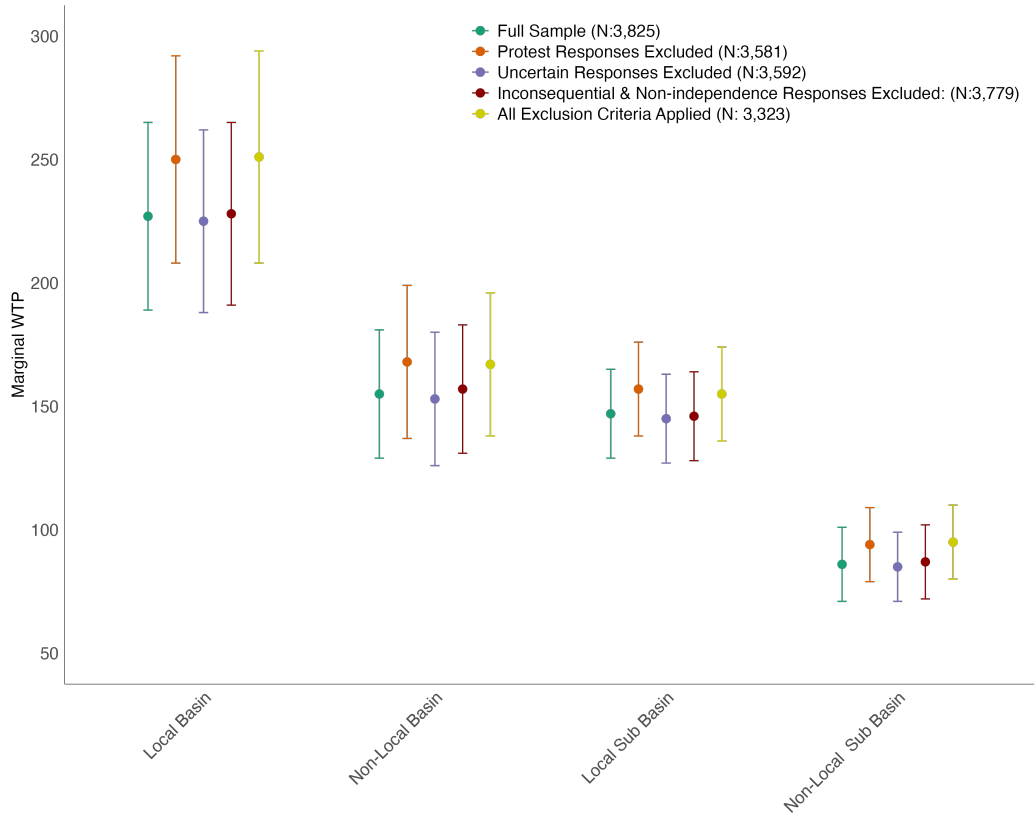


Figure 3.8: Assessing the robustness of model estimates using validity questions.

Notes: Marginal WTP estimates are derived from Model 2, estimated on a restricted sample that excludes respondents based on their answers to sensitivity and consequentiality validity questions, as outlined below:

Protest Responses Excluded: Respondents were excluded if they consistently voted “Yes” or “No” in all choice scenarios and provided responses to follow-up validity questions indicating protest behavior or lack of engagement. This includes: (1) voting “Yes” to all proposals while disagreeing with the claim “I voted as if my household would face the stated cost” and (2) voting “No” to all proposals while agreeing with the statements “I believe that my taxes are too high already and am against any initiative that will increase taxes.”

Uncertain Responses Excluded: Respondents were excluded if they expressed uncertainty in their choices and provided follow-up responses indicating low engagement or questionable reliability. This includes those who disagreed with the statement “I am certain that I voted the same way I would if I were voting in a public election,” while also indicating that (1) “I did not read the information on the proposals carefully” or (2) “I did not have enough information to make comfortable decisions.”

Inconsequential and Non-independence Responses Excluded: Respondents were excluded if they disagreed with statements affirming the survey’s consequentiality. Specifically, exclusions applied to respondents who disagreed with the following statements: “I voted as if the policies would achieve the stated improvements in water quality”, “I voted as if the information collected in this survey will be used to inform policy makers” and “I voted on each proposal without any consideration of the other proposals.”

All Exclusion Criteria Applied: Respondents meeting any of the exclusion criteria described in Criteria 1 through 4 were excluded from the sample.

3.7. Discussion and Conclusion

In SP surveys, particularly those valuing the economic benefits of water quality improvements, defining the SQ water quality is often challenging due to limited data. Consequently, researchers frequently need to make arbitrary decisions about how to frame SQ water quality levels. We examine whether these decisions on arbitrarily defined SQ levels influence public preferences, using a split-sample approach to experimentally manipulate SQ conditions. By leveraging this exogenous variation, we assess how the SQ affects public preferences for water quality improvements. Our use of experimentally induced variation allows for a more robust causal identification of the SQ’s effects on public preferences, overcoming concerns associated with confounding from observational variability in the SQ. We find that researchers’ decisions regarding how the SQ is defined are not merely a technical detail of survey design—they meaningfully shape public preferences for water quality improvements. In particular, people derive higher utility from proposed improvements when baseline water quality is lower, whereas the same improvements are valued less when the SQ is relatively high. Ignoring such perceptual and framing biases when designing SP surveys may misrepresent the true societal value of environmental improvements. Therefore, the careful definition and communication of SQ conditions are critical for producing credible, policy-relevant valuation evidence.

Our empirical findings on public preferences for water quality improvements that inform policy, emphasizing the roles of spatial scale, location, and status quo conditions in determining WTP. Notably, spatial scale drives substantial preference variations, with marginal WTP estimated at local basins and sub-basins, is estimated at \$226, dropping to \$156 for non-local basins. In contrast, sub-basin-level improvements are estimated at \$147 locally versus \$86 non-locally. The total WTP for improving the water quality from each sub-basin’s current baseline to “good” (Level 2) at the sub-basin scale, considering only local values, ranges from \$320 to \$998, with higher WTP corresponding to areas with lower baseline health scores in our study region. WTP patterns across spatial scales and local versus non-local contexts are consistent with the findings of Vossler et al. (2023). Although their spatial scales and water quality scales differ, at their smallest scale, they estimate a marginal WTP of \$121 (2021 USD) for a one-unit increase in their six-level water quality score in a local sub-watershed. At the watershed scale, the marginal WTP for a local watershed is \$164, while it is lower, at \$135 (2021 USD), for a non-local watershed.

There are several limitations of this study. One limitation is arbitrary assignment of water quality values to sub-basins with missing data. Despite employing two scenarios to address uncertainty, this data imputation may affect the observed systematic variation in WTP across baseline conditions. This approach, particularly

relevant in Canada’s data-scarce context, warrants careful scrutiny in future research to ensure robust valuation. Another limitation is the reliance on an online panel sample, which may underrepresent non-internet users, provide insufficient observations for severely degraded subsets, and depend on self-reported recreational behavior, potentially introducing bias. To address this, we applied weights to survey responses to align with 2021 Census population targets, mitigating bias from underrepresented groups.

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